

S I M M Raton Mondol

POST-DOCTORAL FELLOW · INFORMATION AND ELECTRONICS JOINT RESEARCH INSTITUTE

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Profile

Postdoctoral researcher with a Ph.D. in Electrical and Computer Engineering, specializing in biomedical informatics, speech processing, and AI/ML for healthcare. Experienced in applying ML to clinical and speech datasets, especially for Parkinson's disease. Skilled in building classification pipelines integrating acoustic features and ASR outputs. Passionate about developing scalable informatics tools for data-driven clinical outcomes.

Education

Ph.D. in Electrical and Computer Engineering

Sep. 2017 – Aug. 2024

INHA UNIVERSITY, INCHEON, KOREA

M.Eng. in Information and Communication Engineering

Sep. 2013 – Jun. 2016

CHONGQING UNIVERSITY OF POSTS AND TELECOMMUNICATIONS, CHONGQING, CHINA

B.Sc. in Electronics and Telecommunication Engineering

May 2007 – Feb. 2012

DAFFODIL INTERNATIONAL UNIVERSITY, DHAKA, BANGLADESH

Technical Skills

- **Programming Languages:** Python, MATLAB
- **Libraries & Frameworks:** PyTorch, Scikit-learn, TensorFlow, Pandas, Keras, NumPy, SciPy
- **Machine Learning Models:** SVM, RF, MLP, CNN, DNN
- **Speech/Clinical Tools:** Whisper, Wav2Vec 2.0, COVAREP, Praat, Parselmouth

Research Experience

Postdoctoral Fellow

Sep. 2024 – Present

INFORMATION AND ELECTRONICS JOINT RESEARCH INSTITUTE, INHA UNIVERSITY

- Advancing prior doctoral research by developing next-generation AI models for Parkinson's severity staging using enhanced acoustic-ASR feature fusion.
- Leading experiments on scalable, multimodal diagnostic pipelines that integrate speech features with clinical data for improved precision.

Graduate Research Assistant

Sep. 2017 – Aug. 2024

DEPT. OF ELECTRICAL AND COMPUTER ENGINEERING, INHA UNIVERSITY

- Development of machine learning pipelines for multistage classification of Parkinson's disease using acoustic features and ASR outputs, including Whisper fine-tuning.
- Contributed to clinical applications of ML through personalized hearing aid fitting and speech recognition optimization for Korean cochlear implant patients.

Graduate Research Assistant

Sep. 2014 – Jun. 2016

DEPT. OF INFORMATION AND COMMUNICATION ENGINEERING, CQUP, CHINA

- Implemented impulse response-based acoustic echo cancellation systems using speech and random signals.
- Contributed to research on double-talk detection, and voice activity detection for acoustic echo cancellation.

Teaching & Academic Service

Teaching Assistant

Mar. 2022 – Aug. 2022

COLLEGE OF SOFTWARE CONVERGENCE, INHA UNIVERSITY

International Student Ambassador

Mar. 2021 – Aug. 2024

INTERNATIONAL CENTER, INHA UNIVERSITY

Professional Experience

System Administrator

Apr. 2012 – Jul. 2013

ICE TECHNOLOGIES LTD., DHAKA, BANGLADESH

Assistant System Engineer

Apr. 2011 – Mar. 2012

TRIANGLE SERVICES LTD., DHAKA, BANGLADESH

Publications

JOURNALS

1. **Mondol, S.**, Kim, R., Lee, S. (2025). "Advanced optimization strategies for combining acoustic features and speech recognition error rates in multi-stage classification of Parkinson's disease severity." *Biomed. Eng. Lett.*, 15(3), 497. (ISSN: 2093-985X, I.F. = 3.2) DOI: 10.1007/s13534-025-00465-9.
2. **Mondol, S.**, Kim, R., Lee, S. (2023). "Hybrid Machine Learning Framework for Multistage Parkinson's Disease Classification Using Acoustic Features of Sustained Korean Vowels." *Bioengineering*, 10(8), 984. (ISSN: 2306-5354, I.F. = 3.8) DOI: 10.3390/bioengineering10080984.
3. **Mondol, S.**, Kim, H. J., Kim, K. S., Lee, S. (2022). "Machine Learning-Based Hearing Aid Fitting Personalization Using Clinical Fitting Data." *J. Healthc. Eng.*, 2022, 1667672. (ISSN: 2040-2309, I.F. = 3.05) DOI: 10.1155/2022/1667672.
4. Jeong, J., **Mondol, S.**, Kim, Y. W., Lee, S. (2021). "An Effective Learning Method for Automatic Speech Recognition in Korean CI Patients' Speech." *Electronics*, 10(7), 807. (ISSN: 2079-9292, I.F. = 2.6) DOI: 10.3390/electronics10070807.
5. **Mondol, S.**, Lee, S. (2019). "A Machine Learning Approach to Fitting Prescription for Hearing Aids." *Electronics*, 8(7), 736. (ISSN: 2079-9292, I.F. = 2.6) DOI: 10.3390/electronics8070736.
6. Souleymane, B., **Mondol, S.**, Wei, Z. (2016). "Comparative Performance Study of LS and MMSE Channel Estimation over Time Varying Channel in OFDM System." *International Journal of Engineering Research & Technology*, 5(3), 847. (ISSN: 2278-0181)
7. **Mondol, S.**, Zhou, Y. (2014). "Acoustic Echo Cancellation for Speech and Random Signal Using Estimated Impulse Responses." *International Journal of Recent Development in Engineering and Technology*, 2(3), 105. (ISSN: 2347-6435)
8. Haque, A. K. M. F., **Mondol, S.**, Muzahid, A. A. M., Islam, M. Z. (2013). "Acoustic Echo Cancellation for the Advancement in Telecommunication." *International Journal of Advance Innovations, Thoughts & Ideas*, 2(1), 1. (ISSN: 2277-1891, I.F. = 0.83)

CONFERENCES

1. **Mondol, S.**, Yang, Y. J., Cho, J. R., Maeng, S., Lee, S. (2025). "Comparison of Key Acoustic Features of Voices of Patients with Depression and Parkinson's Disease." In *Proceedings of the 2025 Summer Conference of the Korean Society of Medical and Biological Engineering*, Jeju, Korea.
2. **Mondol, S.**, Jeon, J. H., Cho, J. R., Lee, S. (2024). "Stage-Specific Performance of Whisper ASR Model for Parkinson's Speech." In *Proceedings of the 2024 Fall Conference of the Korean Society of Medical and Biological Engineering*, Seoul, Korea.
3. **Mondol, S.**, Kim, R., Lee, S. (2023). "Data Valuation for the Small-Scale Parkinson's Speech Data Using Shapley Value." In *Proceedings of the 2023 Fall Conference of the Korean Society of Medical and Biological Engineering*, Seoul, Korea.
4. Bak, H., Cho, W., Kim, Y., **Mondol, S.**, Jeong, J., Kim, R., Lee, S. (2022). "Speech Recognition on Parkinson's Patients Based on Wav2Vec 2.0." In *Proceedings of the 2022 Summer Conference of the Korean Institute of Electrical and Electronic Engineers*, Jeju, Korea.
5. Bak, H., Cho, W., Kim, Y., **Mondol, S.**, Lee, S. (2022). "Conformer-CTC Based Korean Speech Recognition." In *Proceedings of the 2022 Winter Conference of the Korean Institute of Communications and Information Sciences (KICS)*, Pyeongchang, Korea.
6. **Mondol, S.**, Kim, S., Jeong, J., Lee, S. (2020). "Acoustic Feature Analysis of Speech for Depression Recognition." In *Proceedings of the 2020 Fall Conference of the Korean Multimedia Society*, Yeosu, Korea.
7. **Mondol, S.**, Lee, S. (2019). "Hearing Aid Fitting Data Analysis." In *Proceedings of the 2019 Fall Conference of the Korean Society of Medical and Biological Engineering*, Incheon, Korea.
8. Muzahid, A. A. M., Ingrid, K. M. R., **Mondol, S.**, Zhou, Y. (2016). "Advanced Double-Talk Detection Algorithm Based on Joint Signal Energy and Cross-Correlation Estimation." In *Proceedings of the 2016 8th IEEE International Conference on Communication Software and Networks*, Beijing, China.
9. Muzahid, A. A. M., **Mondol, S.**, Ingrid, K. M. R., Zhou, Y. (2016). "An Efficient Switching Mechanism Using Voice Activity Detection for Acoustic Echo Cancellation." In *Proceedings of the 4th International Conference on Machinery, Materials and Computing Technology*, Beijing, China.
10. **Mondol, S.**, Chung, B. Q., Qi, Z. T. (2015). "Real Orthogonal STBC MC-CDMA Blind Recognition Based on DEM-Sparse Component Analysis." In *Proceedings of the 2015 International Conference on Intelligence Science and Big Data Engineering*, Suzhou, China.

Awards & Fellowships

Outstanding Research Award GRADUATE SCHOOL, INHA UNIVERSITY	2024
Jungseok International Scholarship (100%) GRADUATE SCHOOL, INHA UNIVERSITY	2017
Excellent Masters Thesis Award INTERNATIONAL SCHOOL, CHONGQING UNIVERSITY OF POSTS AND TELECOMMUNICATIONS	2016
Chinese Government Scholarship (100%) CHINESE GOVERNMENT THROUGH BANGLADESH MINISTRY OF EDUCATION	2013
Inter-University ICT Quiz Competition (2nd Prize) DAFFODIL INTERNATIONAL UNIVERSITY	2009

References

References are available upon request